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Comparative stomatal studies in two varieties of *Urginea indica* Kunth collected from Ranchi, Jharkhand

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Abstract : *Urginea indica* Kunth is a plant with high therapeutic values, and has long been used as a source of natural product with pharmaceutical and biocidal applications. In the present investigation comparative stomatal studies in two varieties of Urginea indica Kunth collected from Ranchi, Jharkhand are presented. Anomocytic types of stomata were observed in both the varieties of Urginea indica Kunth. Maximum stomatal index and stomatal size were recorded in V1 variety i.e. in White bulb variety of *Urginea indica* Kunth.

Keywords: *Urginea indica* Kunth, Stomatal index, Anomocytic Stomata

INTRODUCTION

Urginea indica Kunth is a perennial wild herb of Liliaceae family.In India, it occurs both in Southern and in Peninsular part including the coastal as well as in temperate regions of Himalayas.Urginea indica Kunth is considered to have high medicinal value and is largely used as an expectorant, cardiac stimulant and as an anticancer agent. This herb is very unique in its distribution and morphological features. They differ in both vegetative and floral morphology, showing differences in shape, size and color of bulb and leaf, the length of inflorescence and flower color.1The aim of this study, therefore, is to determine variations in leaf epidermal anatomy of selected two varieties of Urginea indica Kunth as influenced by environmental conditions, and also using the epidermal studies in establishing the taxonomic relationship between the two varieties of Urginea indica Kunth. The leaf surface characters can be used as secondary or supporting character in biosystematics studies.[2]

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MATERIALAND METHODS

Two varieties of Urginea indica Kunth were collected from Birsa Agriculture University, Ranchi, Jharkhand. The two varieties of Urginea indica Kunth were distinguished on the basis of color of bulbs, as:

a. Urginea indica Kunth White Bulb Variety -V1
b. Urginea indica Kunth White Bulb Variety -V2

Stomatal studies were done by usual method of mechanical peeling of the epidermis from fresh leaves. Peeling of the epidermis was done at apex, middle and base portion on dorsal and ventral surface of leaf. The epidermis peels were stained with aqueous safranin and mounted in glycerin. The stomatal index was calculated by counting the number of stomata per unit area and dividing it by the sum of number of epidermal cells per unit area and the number of stomata per unit area. Length and width of stomata were measured with the help of ocular and stage micrometer.

$$\text{Stomatal Index} = \frac{\text{No. of Stomata per Unit Area}}{\text{No. of epidermal cells per unit area} + \text{No. of stomata per unit area}} \times 100$$

RESULT AND DISCUSSION:

In the present investigation,the two varieties of Urginea indica Kunth were taken for the comparative study of epidermal anatomical characteristics, such as stomatal index and size of stomata. The data for above stomatal

studies have been depicted in Table 1. The stomatal index was conducted at apex, middle and base portion of both dorsal and ventral surfaces.

Table1:Stomatal index(%), Length and Width(in µ) of stomata in Dorsal and ventral surface of the two varieties of Urginea indica Kunth.

SURFACE OF LEAVES	VARIETIES	Apex Portion of Leaf			Middle Portion of Leaf			Base Portion Of Leaf		
		Stomatal Index	Length (µ)	Width (µ)	Stomatal Index	Length (µ)	Width (µ)	Stomatal Index	Length (µ)	Width (µ)
DORSAL SURFACE	V_1	10.83 ± 0.404	19.5 ± 0.092	13.1 ± 0.049	14.221 ± 0.309	18.5 ± 0.128	15.3 ± 0.118	8.696 ± 0.304	19.0 ± 0.289	14.3 ± 0.173
	V_2	11.573 ± 0.330	19.5 ± 0.095	13.1 ± 0.071	10.564 ± 0.248	17.4 ± 0.190	14.5 ± 0.111	6.090 ± 0.188	18.9 ± 0.137	13.8 ± 0.132
VENTRAL SURFACE	V_1	14.933 ± 0.331	19.9 ± 0.094	**17.0 ± 0.011**	**19.801 ± 0.328**	19.0 ± 0.154	16.1 ± 0.094	8.755 ± 0.302	**20.5 ± 0.067**	16.2 ± 0.097
	V_2	11.766 ± 0.271	17.8 ± 0.146	10.6 ± 0.101	11.122 ± 0.247	20.6 ± 0.091	12.00 ± 0.089	6.576 ± 0.193	20.10 ± 0.030	10.6 ± 0.101

V_1: white Bulb variety of Urginea indica Kunth.
V_2: Brown Bulb variety of Urginea indica Kunth.

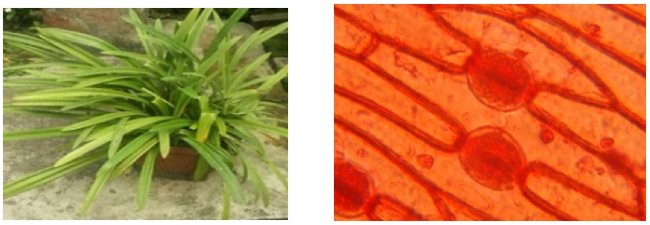


FIG1: Leaves ofUrginea indica Kunth White Bulb Variety (V1) with Photomicrograph showing its Stomata.



FIG2: Leaves of Urginea indica Kunth Brown Bulb Variety (V2) with Photomicrograph showing its Stomata.

Various studies have emphasized the significance of foliar epidermal anatomy in the systematics of many plant groups,3-5 thus suggesting that leaf surface patterns, anatomical features and structures are genetically controlled.

The leaves of both the varieties of Urginea indica Kunth are long, lanceolate, pointed but the leaves of White Bulb Variety (V1) of Urginea indica Kunth were comparatively broader in shape and lighter in color (Fig. 1 and 2). All the stomata of both the varieties were of anomocytic type with no subsidiary cells. Each stoma was surrounded by two kidney shaped guard cells, of which the middle portion was narrow. There were slight difference in all the parameters i.e. stomatal index, stomatal length and stomatal width of the leaf on both dorsal and ventral surfaces. Stomatal index and stomatal size of White Bulb Variety (V1) of Urginea indica Kunth were comparatively higher. Maximum stomatal index (19.801±0.328) was recorded at the middle portion on the ventral surface of White Bulb Variety (V1) of Urginea indica Kunth. There were slight differences in the length and width of stomata of both the varieties of Urginea indica Kunth. Highest length recorded was 20.500±0.067µ at base portion of the leaf on dorsal surface and highest width was 17.00±0.011µ at the apex portion on the ventral surface of White Bulb Variety (V1) of Urginea indica Kunth.

The result shows the difference between the two varieties of Urginea indica Kunth. Epidermal characters, that is, epidermal cells and stomata have proven to be an important tool in distinguishing medicinal plants.6 Stomatal function is important beyond the level of plant physiology and function, and its significance reaches from evolutionary history to atmospheric and environmental sciences. Stomatal size is positively correlated with genome size across a wide range of major taxa. Stomatal size predicts genome size within angiosperms.7This investigation concludes that stomatal variations in the two varieties of Urginea indica Kunth are of taxonomic significance.

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REFERENCES:

1. Shiva Kameshwari, M.N.; Tharasaraswathi, K.J. and Muniyamma, M.2010. Morphological variations in populations of Urginea indica Kunth. Liliaceae. Journal of Applied and Natural Science. 2(2): 280-289.
2. Shiva Kameshwari, M. N.; BijulLakshman, A. and Paramasivam G.2012. Biosystematicstudies on medicinal plant Urginea indica Kunth. Liliaceae-A review. Int. Journ. Pharm. Life Sci 3(1).
3. Wilkins P.W. and Sabanci C.O. 1990. Genetic variation in leaf epidermal cell size and shape in Loliumperenne.Euphytica. 47:233-239.
4. Mustapha O.T.2000. Cytotaxonomy of the genus Urgineastein III: The taxonomic value of foliar anatomical features inUrginea indica(Roxb.)Kunth. Complex. Biosci. Res. Comm. 12(2):201-206.

5. Ahmad, K.; Khan, M.R.; Ahmad, M.; Shaheen, N.; Nazir, A.2010. Taxonomic diversity in epidermal cells of some sub-tropical plant species. Int. Journal Agric. Biol. 12:115-118.

6. Johr, A. June 2013. Epidermal characters of some ornamental plants used in Traditional medicine. Advances in plant science. 26(1):153-155.

7. Hodgson, J.G.; Sharafi, M.; Jalil, A.; Diaz, S. April, 2010. Stomatal vs. genome size in angiosperms: the somatic tail wagging the genome DOG. Annals of Botany. 105(4): 573-584.

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